CS1202 — Discrete Mathematics II (graphs, algebra, number theory)

Tier	A
Phase (order)	2
Do this first	CS1200
Estimated effort	8-10 weeks, ~5 h/week
Unlocks	CS2320, CS2800, CS6111 (cryptography), CS6030

Roadmap

1. Graph theory basics: paths, cycles, trees, bipartite, Euler/Hamilton, connectivity, Menger, matchings, planarity.
2. Algebraic structures: groups, rings, fields.
3. Number theory: gcd, Bezout, modular arithmetic, CRT, Fermat.

Video courses (all links checked on YouTube on 2026-10-22)

🎓 Main course: follow it start to end

- MIT 6.042J Mathematics for Computer Science, Fall 2010 (25 videos). MIT 6.042J: number theory (lectures on GCD/modular arithmetic/RSA), graphs, and counting.

🎥 Other full lecture series (pick one if the main one doesn't click, or for topics it skips)

- Computer - Graph Theory (40 videos). NPTEL *Graph Theory* (nptelhrd).
- Discrete Mathematics - IIITB (61 videos). NPTEL-NOC IITM *Discrete Mathematics* (IIIT Bangalore): covers algebraic structures and number theory.

⚡ Short videos for intuition and revision

- Number theory Tutorials | Competitive Programming (25 videos). Luv *Number theory*: applied, competitive-programming style.

Study plan (week by week)

Lecture numbers are positions in the main playlist above. Click a number to open that video.

Week	Lectures	What to do
1-3	MIT 6.042 lectures on number theory (GCD, modular arithmetic, RSA)	Extended Euclid, modular inverse, CRT, Fermat/Euler; implement RSA toy in Python.

Week	Lectures	What to do
4-6	MIT 6.042 graph lectures + NPTEL Graph Theory (alt)	Graph basics, connectivity, trees, matchings, colouring. Prove Hall's theorem.
7-9	NPTEL IIIT-B Discrete Maths (alt) algebra part	Groups, rings, fields, lattices. Do Rosen/Gallian exercises.
10	MIT 6.042 counting lectures	Advanced counting + generating functions. Solve 15 Project-Euler-style counting problems.

▼ **All lectures in *MIT 6.042J Mathematics for Computer Science, Fall 2010* (25 videos)**

1. Lec 1 | MIT 6.042J Mathematics for Computer Science, Fall 2010
2. Lec 2 | MIT 6.042J Mathematics for Computer Science, Fall 2010
3. Lec 3 | MIT 6.042J Mathematics for Computer Science, Fall 2010
4. Lec 4 | MIT 6.042J Mathematics for Computer Science, Fall 2010
5. Lec 5 | MIT 6.042J Mathematics for Computer Science, Fall 2010
6. Lec 6 | MIT 6.042J Mathematics for Computer Science, Fall 2010
7. Lec 7 | MIT 6.042J Mathematics for Computer Science, Fall 2010
8. Lec 8 | MIT 6.042J Mathematics for Computer Science, Fall 2010
9. Lec 9 | MIT 6.042J Mathematics for Computer Science, Fall 2010
10. Lec 10 | MIT 6.042J Mathematics for Computer Science, Fall 2010
11. Lec 11 | MIT 6.042J Mathematics for Computer Science, Fall 2010
12. Lec 12 | MIT 6.042J Mathematics for Computer Science, Fall 2010
13. Lec 13 | MIT 6.042J Mathematics for Computer Science, Fall 2010
14. Lec 14 | MIT 6.042J Mathematics for Computer Science, Fall 2010
15. Lec 15 | MIT 6.042J Mathematics for Computer Science, Fall 2010
16. Lec 16 | MIT 6.042J Mathematics for Computer Science, Fall 2010
17. Lec 17 | MIT 6.042J Mathematics for Computer Science, Fall 2010
18. Lec 18 | MIT 6.042J Mathematics for Computer Science, Fall 2010
19. Lec 19 | MIT 6.042J Mathematics for Computer Science, Fall 2010
20. Lec 20 | MIT 6.042J Mathematics for Computer Science, Fall 2010
21. Lec 21 | MIT 6.042J Mathematics for Computer Science, Fall 2010
22. Lec 22 | MIT 6.042J Mathematics for Computer Science, Fall 2010
23. Lec 23 | MIT 6.042J Mathematics for Computer Science, Fall 2010
24. Lec 24 | MIT 6.042J Mathematics for Computer Science, Fall 2010
25. Lec 25 | MIT 6.042J Mathematics for Computer Science, Fall 2010

Books, notes and websites

- Same NPTEL discrete-math courses as CS1200 (second half)
- MIT 6.042J (graphs, number theory chapters)
- Book: Bondy & Murty (graph theory) or West, *Introduction to Graph Theory*
- Book: *A Computational Introduction to Number Theory and Algebra* (Victor Shoup, free PDF)

Practice / proof of understanding

Prove Euler's characterisation, Hall's theorem and Fermat's little theorem in your own words. Implement extended Euclid and fast modular exponentiation.

What your IITM course sheet says

From the PDF: CS1202

Official description: ? Basics of Graph Theory (8 Weeks) Basic definitions & isomorphism, walks, trails, paths, cycles, forests & trees, connected vs disconnected graphs, subgraphs, spanning subgraphs & spanning trees, bipartite graphs vs odd cycles (with proof), vertex colourings, Eulerian tours vs Hamiltonian cycles, characterisation of Eulerian graphs (with proof), vertex-connectivity & edge-connectivity, Menger’s Theorem (with proof), Matchings and Halls Theorem. Planar graphs, Euler formula. Digraphs, DAGs, Tournaments - basics and properties ? Algebraic Structures (4 weeks) Semigroups, Monoid and Groups, Rings, and Fields. Vector spaces, bases. ? Basic Number Theory (2 Weeks) Divisibility, GCD, Bezout’s Lemma, Properties of primes. Modular arithmetic, Congruences, Chinese remainder theorem. Fermat's little theorem.

Official syllabus (trimmed):

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Official textbooks: Text Books 1. Graph Theory by Bondy and Murty, Springer 2008 2. Abstract Algebra, by I. N Herstein, 3rd edition, Prentice-Hall, 1996 3. An Introduction to the Theory of Numbers, Ivan Niven, Herbert S. Zuckerman and Hugh Montgomery, 5th Edition, John Wiley & Sons, 1991

Official references: Reference Books 1. Introduction to Graph Theory by Douglas B. West, Second Edition, Pearson, 2006 2. Mathematics for Computer Science Eric Lehman, F Tom Leighton and Albert R Meyer - https://courses.csail.mit.edu/6.042/spring18/mcs.pdf 3. Building blocks for TCS (Fleck) - https://mfleck.cs.illinois.edu/building-blocks/index-sp2020.html